

OLA Operations & Revenue Dashboard: Bengaluru City

Project Overview

Objective: The primary goal of this project was to analyze operational efficiency and revenue generation for OLA's ride-hailing network in Bengaluru. By examining a robust dataset of daily rides, this analysis identifies critical trends in booking success rates, vehicle category performance, and the underlying causes of ride cancellations by both drivers and customers.

Project Scope & Dataset

- **Location:** Bengaluru City.
- **Timeframe:** One month of operational data.
- **Data Volume:** 100,000 individual ride records (1 lac rows) spanning 21 distinct variables.
- **Key Metrics Analyzed:** Booking status (targeting a ~62% success rate baseline), average ride distances, booking values, payment methods, and customer/driver ratings.

Technical Stack

- **Data Cleaning & Preparation:** Python (Pandas) / Power Query (Addressed missing values, standardized formatting, and removed duplicate primary keys).
- **Database Management & Querying:** Microsoft SQL Server (SSMS) utilizing T-SQL to extract key performance metrics and create targeted data views.
- **Data Visualization:** Power BI (Designed an interactive dashboard to visualize revenue distribution, cancellation bottlenecks, and ride volume trends).

Business Value This comprehensive analysis transitions raw transactional data into actionable strategic insights, providing a clear window into operational bottlenecks (such as driver-initiated cancellations) and revenue-driving factors (such as peak booking times and dominant payment methods).

Dataset Summery:

Overview The analysis is built upon a synthesized dataset containing 1,03,022 individual ride records for OLA operations within Bengaluru city. The data captures a comprehensive one-month operational window, providing a robust sample size for identifying trends in user behavior, driver performance, and financial outcomes.

Data Structure & Size

- **Total Records:** 1,03,022 rows (individual ride bookings).
- **Total Variables:** 21 columns.
- **Geographic Scope:** 50 distinct pickup and drop-off locations across Bengaluru.
- **Target Success Baseline:** The dataset inherently reflects an overall successful booking rate of approximately 62%.

Key Data Categories

- **Time & Location:** Date and time of the booking, alongside specific pickup and drop-off coordinates.
- **Identifiers:** Unique 10-digit alphanumeric alphanumeric booking IDs (e.g., CNR...) and customer IDs.
- **Vehicle Fleet:** Categorized into 7 specific ride types (Auto, Prime Plus, Prime Sedan, Mini, Bike, eBike, and Prime SUV).
- **Performance Metrics:** Vehicle Turnaround Time (V_TAT), Customer Turnaround Time (C_TAT), Driver Ratings, and Customer Ratings.
- **Financials:** Total estimated or collected booking value and the associated payment method (Cash, UPI, Credit Card).
- **Status & Friction Points:** Categorical flags indicating if a ride was successful, canceled by the driver, canceled by the customer, or incomplete, complete with specific reasons for failure (e.g., "Customer Demand," "Vehicle Breakdown").

Exploratory data analysis using Excel and python

Data Cleaning & Transformation

Overview The raw dataset required significant preprocessing to ensure accuracy and consistency before executing SQL queries or developing the Power BI dashboard. All data manipulation and structural formatting were handled utilizing Microsoft Excel.

Key Transformations

- **Date & Time Repair:** Addressed corrupted timestamp strings (e.g., "26-7-224 2:: PM") by utilizing Text-to-Columns and Find & Replace functions to isolate the date, correct the year formatting to 2024, and standardize the time into a clean HH:MM:SS format.
- **Text Standardization:** Enforced lowercase formatting across all 21 column headers and categorical row data to prevent syntax errors during the SQL database import.
- **Error & Null Handling:** Completely removed corrupted columns containing #NAME? formula errors and converted literal text strings of "null" into true blank values to allow for accurate numeric aggregations.
- **Data Integrity Checks:** Identified and removed duplicate booking_id records to ensure the column could act as a strict Primary Key in the SQL database.
- **Logical Filtering:** Evaluated canceled rides (by both driver and customer) and cleared out residual estimated booking_value data, ensuring that total revenue calculations only reflected successfully completed rides.

Data Issue	Raw Data Example (Before)	Cleaned Data (After)	Excel Method Applied

Corrupted Timestamps	26-7-224 2:: PM	2024-07-26 14:00:00	Text-to-Columns, Find & Replace
Inconsistent Headers	Booking_Status	booking_status	Lowercase function / Manual override
Formula Errors	#NAME?	(Column Removed)	Column Deletion
False Text Nulls	"null"	(True Blank Cell)	Find & Replace
Illogical Revenue	444 (on canceled ride)	(Blank)	Conditional Filtering

```

Cleaning_bookings.py X ola_data_ready_for_sql.csv
Cleaning_bookings.py
1 import pandas as pd
2
3 # Load your dataset
4 df = pd.read_csv('ola_data_final.csv')
5
6 # See how many duplicates you actually have
7 duplicate_count = df.duplicated(subset=['booking_id']).sum()
8 print(f"Found {duplicate_count} duplicate booking IDs.")
9
10 # Drop the duplicates (this keeps the first one it finds and deletes the rest)
11 df = df.drop_duplicates(subset=['booking_id'], keep='first')
12
13 # Save the perfectly clean version
14 df.to_csv('ola_data_ready_for_sql.csv', index=False)

```

Data Analysis Using SQL (Business Transaction)

1. All successful bookings.

	date	time	booking_id	booking_status	customer_id	vehicle_type	pickup_location	drop_location	v_fat	c_fat	customer_cancellation_reason	driver_cancellation_reason	incomplete_rides	incomplete_rides_rea
1	2024-07-15	20:05:00.0000000	cnr111111475	success	cid14457	ebike	fraser town	sahakar nagar	189	1	NULL	NULL	no	NULL
2	2024-07-08	13:16:00.0000000	cnr1111128727	success	cid976233	prime plus	cox town	chickpet	21	9	NULL	NULL	no	NULL
3	2024-07-29	21:29:00.0000000	cnr111134945	success	cid74415	auto	peenya	hosur road	14	1	NULL	NULL	no	NULL
4	2024-07-16	00:06:00.0000000	cnr1111693834	success	cid8265	bike	banashankari	rajajinagar	189	115	NULL	NULL	no	NULL
5	2024-07-25	23:15:00.0000000	cnr1111945197	success	cid313344	prime plus	jayanagar	kr puram	133	9	NULL	NULL	no	NULL
6	2024-07-01	00:37:00.0000000	cnr111196819	success	cid297733	mini	sahakar nagar	mg road	77	125	NULL	NULL	no	NULL
7	2024-07-12	23:29:00.0000000	cnr1111983549	success	cid118765	prime plus	majestic	whitefield	14	145	NULL	NULL	no	NULL
8	2024-07-26	15:04:00.0000000	cnr111215312	success	cid5263	auto	shantinagar	sarjapur road	245	12	NULL	NULL	no	NULL
9	2024-07-01	00:46:00.0000000	cnr1112394941	success	cid261248	auto	malleswaram	chickpet	77	65	NULL	NULL	no	NULL
10	2024-07-04	23:53:00.0000000	cnr11124614	success	cid51118	ebike	basavanagudi	fraser town	119	12	NULL	NULL	no	NULL
11	2024-07-18	04:07:00.0000000	cnr111254365	success	cid789342	prime plus	jayanagar	rajajinagar	28	6	NULL	NULL	no	NULL
12	2024-07-15	20:41:00.0000000	cnr1112726141	success	cid356838	prime plus	devanahalli	kr puram	196	25	NULL	NULL	no	NULL

2. The average ride distance for each vehicle type.

	Avarage_Distance	vehicle_type
1	23.36	prime plus
2	23.25	ebike
3	23.19	mini
4	23.18	bike
5	23.11	prime sedan
6	23.06	prime suv
7	9.52	auto

3. The total number of cancelled rides by customers.

	Total_Cancellation_By_Customer
1	10497

4. List of the top 5 customers who booked the highest number of rides.

	customer_id	total_rides
1	cid4614	7
2	cid7322	7
3	cid1298	6
4	cid5917	6
5	cid95471	6

5. The number of rides cancelled by drivers due to personal and car-related issues.

	cancelled_by_driver_count
1	6541

6. The maximum and minimum driver ratings for Prime Sedan bookings.

	Max_Rating	Min_Rating	vehicle_type
1	5	3	prime sedan

7. All rides where payment was made using UPI.

	date	time	booking_id	booking_stat...	customer...	vehicle_type	pickup_location	drop_location	v_tat	c_tat
31	2024-07-21	16:14:00.0000000	cnr1118578267	success	cid362825	prime plus	kadugodi	mg road	31	11
32	2024-07-11	07:37:00.0000000	cnr1118636123	success	cid484774	auto	indiranagar	chamarajpet	112	55
33	2024-07-15	01:16:00.0000000	cnr1118726	success	cid743351	bike	kr puram	langford town	21	135
34	2024-07-02	19:00:00.0000000	cnr111898247	success	cid26281	mini	malleshwaram	sarjapur road	91	45
35	2024-07-04	08:17:00.0000000	cnr1119159658	success	cid51266	auto	indiranagar	hosur road	266	7
36	2024-07-03	06:24:00.0000000	cnr1119511	success	cid63292	prime sed...	vijayanagar	kammanahalli	266	115
37	2024-07-07	21:16:00.0000000	cnr1119943747	success	cid9283	mini	varthur	btm layout	63	3
38	2024-07-14	05:24:00.0000000	cnr112117559	success	cid49932	prime suv	whitefield	shivajinagar	175	3

8. The average customer rating per vehicle type.

	Avarage_Customer_Rating	vehicle_type
1	4.01	prime plus
2	4	auto
3	4	mini
4	4	prime sedan
5	4	prime suv
6	3.99	bike
7	3.99	ebike

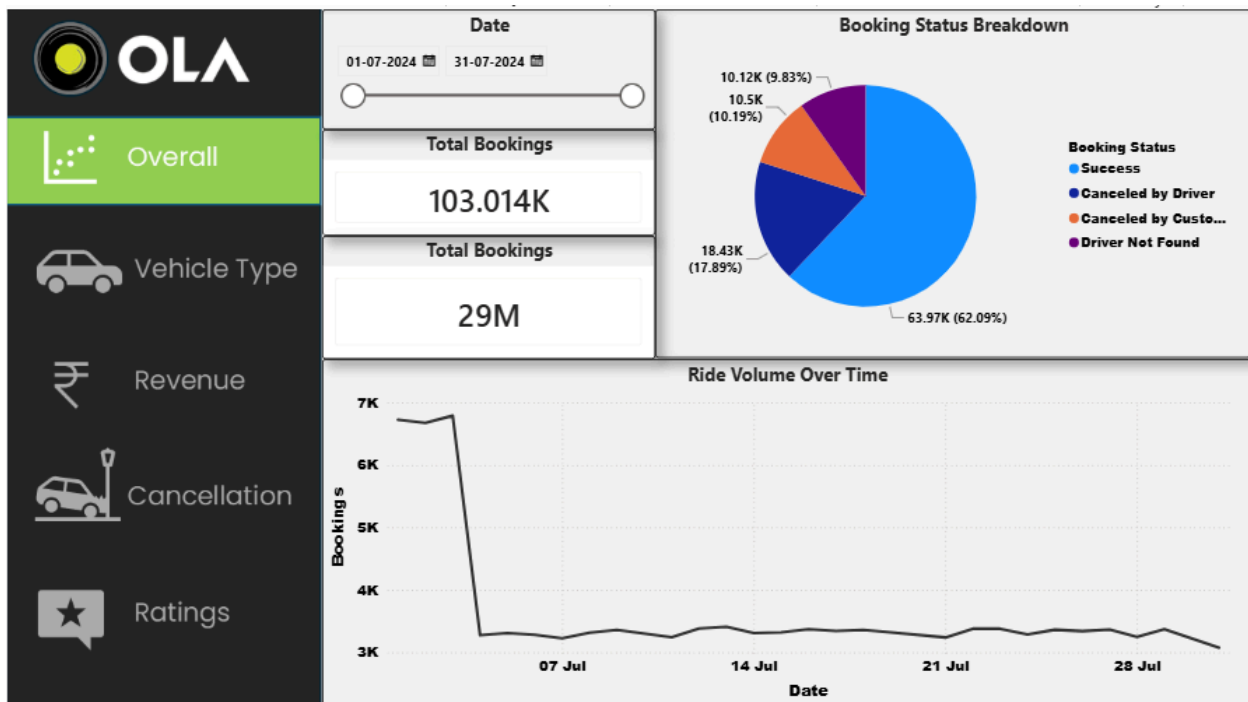
9. Calculation of the total booking value of rides completed successfully.

	Total_Revenue
1	28677287

10. List all incomplete rides along with the reason.

	Total_Incomplete_Rides	incomplete_rides_reason
1	1601	customer demand
2	734	other issue
3	1591	vehicle breakdown

Dashboard using power BI



Overall

Vehicle Type

Revenue

Cancellation

Ratings

Vehicle type

Vehicle Type	Total Booking Value	Success Booking Value	Avg. Distance Travelled	Total Distance Travelled
Prime Sedan	7M	4M	23.11	217K
Prime SUV	6M	4M	23.05	207K
Prime Plus	7M	4M	23.36	212K
Mini	7M	4M	23.19	210K
Auto	7M	4M	9.52	87K
Bike	7M	4M	23.18	212K
E-Bike	7M	4M	23.25	213K

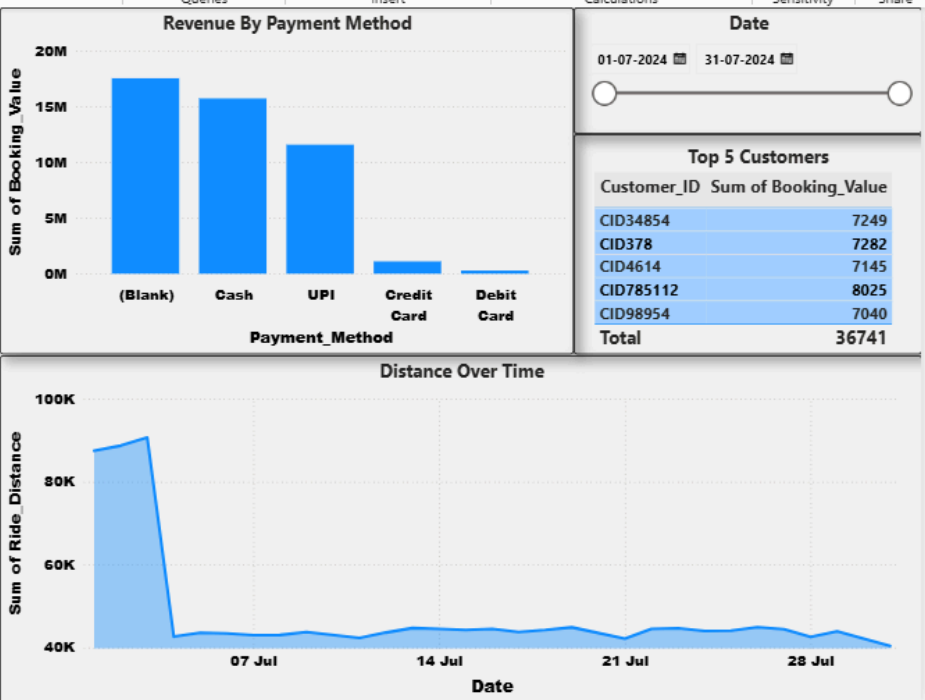
Overall

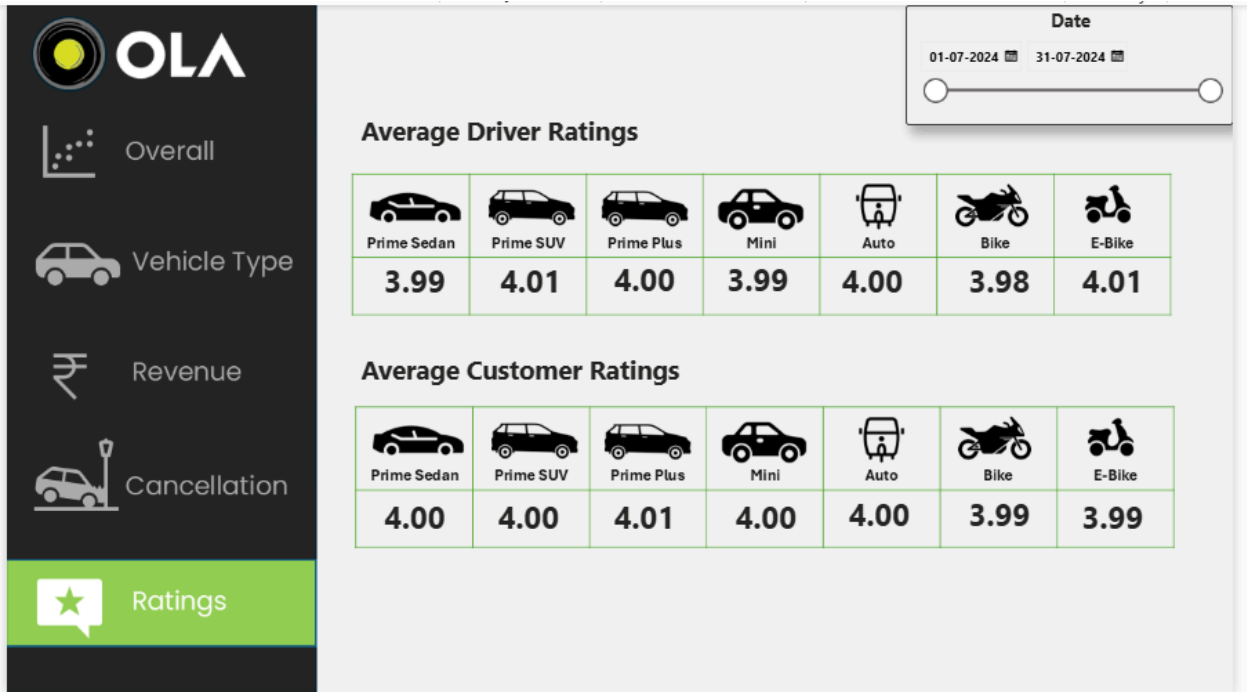
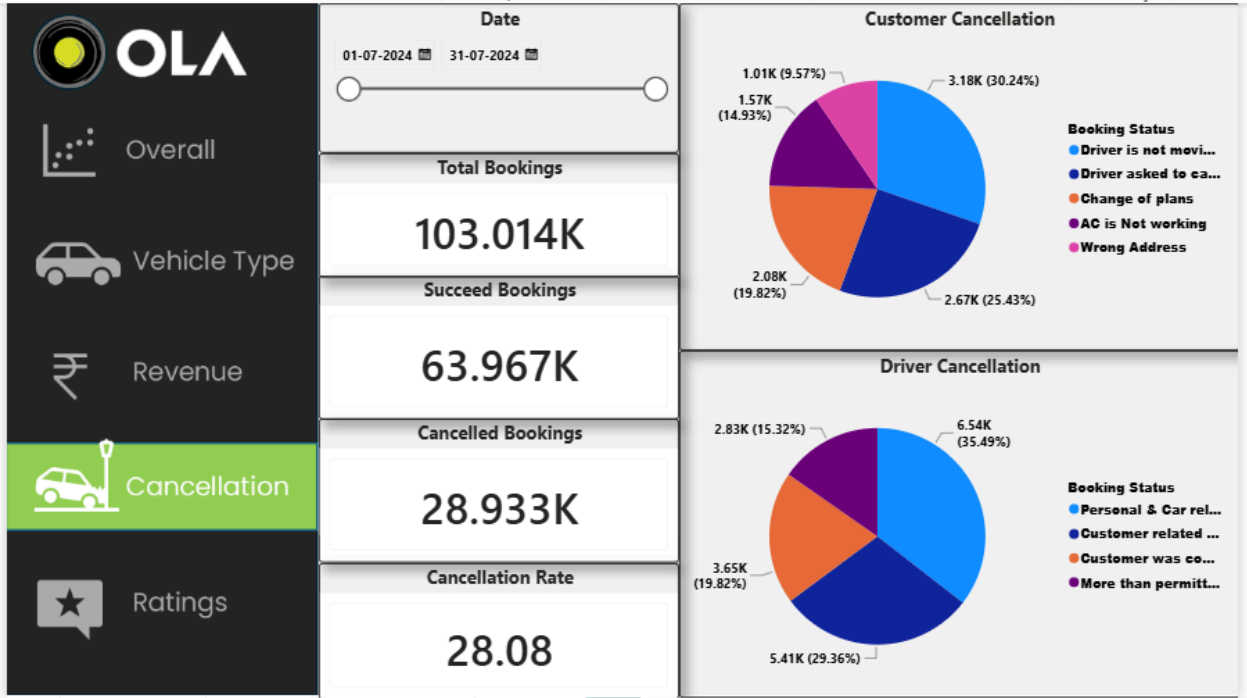
Vehicle Type

Revenue

Cancellation

Ratings





Strategic Business Recommendation

- **Driver Network & Relationship Management:** A significant portion of ride cancellations stems from drivers (e.g., "Personal & Car related issues"). To mitigate this, OLA should implement a structured relationship management program. By treating top-performing, high-rated drivers as key accounts, you can offer them targeted support, such as subsidized vehicle maintenance tie-ups or prioritized routing. Actively managing this network and keeping driver satisfaction high will directly reduce churn and lower the driver-initiated cancellation rate.
- **Mitigating Customer Cancellations:** The data highlights that customers frequently cancel because the "Driver is not moving towards pickup location." OLA should tighten the backend algorithms tracking Vehicle Turnaround Time (V_TAT). If a driver does not move within a 3-minute window of accepting a ride, the system should automatically prompt the driver or auto-reassign the ride to the next closest vehicle to protect the customer experience.
- **Incentivizing Digital Payments:** With a clear view of revenue by payment method, OLA can optimize transaction efficiency. Cash transactions often cause friction and delays. By offering a nominal discount (e.g., 2% off) or loyalty points for rides paid via UPI, OLA can speed up the drop-off process, reduce cash-handling disputes for drivers, and secure revenue faster.
- **Strategic Fleet Rebalancing:** Based on the "Top 5 Vehicle Types by Ride Distance" and "Average Customer Ratings by Vehicle Type" metrics, OLA should evaluate its fleet composition. If eBikes and Prime Sedans consistently yield the highest ratings and longest average distances, the business development team should focus on sourcing more of these specific vehicle types to match market demand.